STAFF REVIEW · H446 092 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. 21.

2. Data/instruction transfer with memory.

3. Control Unit.

4. $1011\ 0000_2 = B0_{16}$.

I Do — source worked steps

1. Explain the mechanism — Address bus12; control bus READ; data bus returns99, placed in MDR. — Address12 and content99 are different types of information.

We Do — guided CHECK

1. Explain the mechanism
Address bus44, data bus25 from CPU to memory, control bus WRITE.
On writing, data flows towards memory while address still identifies destination.

You Do — source mark schemes

1. Address bus7 [2], control read [2], data bus150 [2].

2. Address CPU→memory20 [2], data CPU→memory9 [2], control WRITE [2].

3. PC points instruction address [1], MAR holds address [1], MDR carries instruction contents [2], CIR holds decoded/current instruction [2].

Exit Ticket — source answer

1. False. Address selects location; data represents content.